THE MAGNETIZATION OF THE ROSE HILL FORMATION AT THE CLASSICAL SITE OF GRAHAM'S FOLD TEST

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Abstract. A reinvestigation of the middle Silurian Rose Hill Formation yields a revised pole position of 19.0°N, 129.1°E (α_{95}=5.8° for N=35 samples from nine sites and three lithologies). This is based upon three separate fold tests, all of which indicate predeformational stability of the magnetization at the 99% confidence level, as well as upon normal and reversed polarity directions. Laboratory methods used to derive these results included thermal, chemical, and alternating field stepwise demagnetizations of three lithologies: gray sandstones, hematitic red sandstones, and dolomitic sandstones. Consistency among the techniques was excellent, and through vector subtraction we obtained characteristic and secondary components of magnetization. Reversed polarity directions of the hematite sandstones did not produce a positive fold test, and their resemblance to late Paleozoic directions for North America suggests that they have been remagnetized during that time. The remagnetization was probably a chemical event, since a thermal event would have also remagnetized the gray sandstones and the dolomitic sandstones. In addition to paleomagnetic experiments, rock magnetic experiments were performed to clarify the nature of the dolomites (previously called siderites). Analysis of natural and isothermal remanent magnetizations from -196°C to 680°C shows a dominance of fine-grained magnetite and hematite. Thin sections show that lesser amounts of coarse-grained magnetite and hematite are also present. Consistency between thermal, chemical, and alternating field demagnetizations indicates that both hematite and magnetite carry the characteristic direction. The mean inclination for the Rose Hill Formation, -44°, places North America in moderate southerly latitudes (about 26°) during the Middle Silurian. The declination of 330° indicates a rotation of 30° in a clockwise sense from the present-day orientation, but it is less unambiguous than the inclination, since we cannot determine if relative rotations of blocks in this part of the Appalachians occurred, or whether the declination is truly representative of the entire North American craton. However, evidence for horizontal rotations on the scale of 1 m to 100 km is lacking, so we prefer a rotation of the entire North American craton by this amount.

Introduction

The purpose of this report is to clarify the nature of magnetization and give a revised pole position of the middle Silurian Rose Hill Formation. The Rose Hill is found in the Valley and Ridge Province of the central Appalachians and has been the topic of two previous paleomagnetic research projects.

The first, by Graham [1949], used the Rose Hill to demonstrate his now classic fold test. His study examined the local and regional consistency of its directions of magnetization. From Pinto, Maryland, 20 samples of what he called siderite-cemented sandstone contained in a fold less than 10 m across formed the basis of his local consistency test. Natural remanent magnetization (NRM) directions of these samples showed an improvement in clustering which was significant at the 99% confidence level [McElhinny, 1964] after bedding corrections were applied. Regional consistency was verified by a favorable comparison of these results with samples collected at Great Cacapon Mountain, West Virginia, 60 km east of the Pinto site.

Subsequent work [R. B. French, 1976; and French and Van der Voo, 1977] on 107 samples of red sandstones from the Rose Hill in Virginia and West Virginia showed that thermal and chemical demagnetizations removed a small secondary direction at 350°C, or 150 hours of leaching, revealing in most cases a reversed polarity characteristic direction which subsequently decayed to the origin. Although this reversed polarity direction did not conclusively pass the fold test, there were a few samples with normal polarity components which did pass the fold test. From these 12 samples a tentative Silurian pole of 26°N, 116°E for North America was obtained.

A follow-up study on these previous investigations was desirable for two reasons: first, we wished to determine with present-day techniques why Graham's fold test worked so well with just natural remanent magnetization (NRM) directions, and, second, we wanted to compare the paleomagnetic properties of the redbeds with those from the other lithologies. In conjunction with other Paleozoic poles a Silurian pole from the Rose Hill Formation would refine the North American polar wander path and consequently indicate the relative latitudinal position of North America during the closure of Iapetus.

Sampling and Local Geology

The Rose Hill Formation, part of the Clinton Group, was deposited in the Middle Silurian about 420 m.y. ago and subsequently deformed in Late Carboniferous to Earliest Permian times, about 300 to 270 m.y. ago [Colton, 1970; Van der Voo, 1979]. In Maryland and northern West Virginia, where our sampling was carried out (Figure 1), three lithologic subdivisions may be made [Swartz, 1923]: (1) lower shales and sandstones, 58 m thick, beginning at the top of the Tuscarora sandstone, (2) Cresaptown iron sandstone, 10 m thick, and (3) upper shale beds, 100 m thick, ending at the bottom of the Keefer sandstone.

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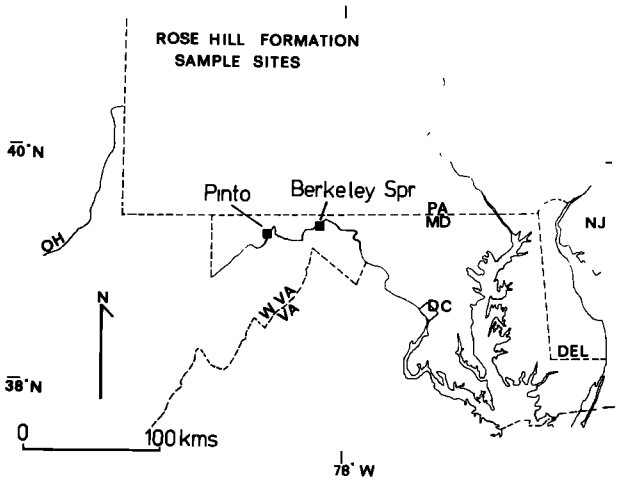


Fig. 1. Map with the two sampling sites of this study.

Three types of rocks appeared to be paleomagnetically suitable at Pinto, Maryland and Great Cacapon Mountain near Berkeley Springs, West Virginia: gray sandstones, red sandstones, and ferroan dolomites (siderites). Gray sandstones are found at the bottom of the lower shales and sandstone sequence, exposed in tight folds at Cacapon Mountain. Representative of a transitional environment from the underlying Tuscarora sandstone to the overlying shales, the gray sandstones have about 5% hematite and accessory amounts of magnetite. Red sandstones from the Cresaptown iron sandstone sequence are exposed at Pinto; here they are not tightly folded and display dips from 15° to 48°. They contain grains of varying texture set in a hematitic matrix. Gradations from oolitic sandstones at Pinto to fine-grained sandstones at Cacapon Mountain indicate a local depositional environment. The ferroan dolomitic sandstones from the upper shales contain accessory amounts of magnetite and traces of hematite. Graham [1949] reported having seen detrital magnetite grains in thin-section studies. Thin sections from some of our samples show bedding-oriented laminae of magnetite grains (as well as other heavy minerals) and therefore also indicate a detrital origin for the magnetite. Lenses of 15-cm thickness and up to 500 m long are dipping 28° at Cacapon Mountain but are more complexly folded at Pinto. Graham's classic fold test is based upon the ferroan dolomites (siderites) from the fold at Pinto, and our sampling localities were selected on the basis of the descriptions and photographs in his paper.

Field and Laboratory Techniques

Our collection sites were located in the section on the basis of descriptions by Swartz [1923], Stose and Swartz [1912], and Graham [1949]. Bedding attitudes were measured with a Brunton compass and wherever possible checked with primary sedimentary features. All red sandstones (43 samples) were drilled in the field and have at most 4° measurement error. From the gray sandstones and dolomitic sandstones, 23 hand samples were taken and drilled in the laboratory; hence they have at most 5° measurement error. Measurements of the magnetization were made with a Super-Conducting Technology $1\frac{1}{2}$" cryogenic magnetometer and a Schonstedt SSM-1A magnetometer. Thermal demagnetizations were done either with a Schonstedt thermal demagnetizer or with a furnace in field-free space provided by Helmholtz coils. Alternating field demagnetizations up to 100 mT were done with a Schonstedt GSD-1 AC demagnetizer. Chemical demagnetizations, using HCl acid in normalities from 6 to 12 at room temperature inside a set of Helmholtz coils, were done on samples which were slotted or cored [Henry, 1979]. A few supplemental measurements utilized equipment at the Institut für Geophysik, ETH-Zürich, and at the Paleomagnetic Laboratory of Utrecht State University. Apparatus used included a thermal demagnetizer which measures continuously while heating [Heiniger and Heller, 1976] and alternating field demagnetizing coils capable of reaching 300 mT [As and Zijderveld, 1958]. Characteristic directions are summarized in Table 1; further details are given by A. N. French [1978].

Results

Gray sandstones. From the tightly folded section of the lower shale sequence at Cacapon Mountain, 10 gray sandstone samples were collected and thermally demagnetized. Contrary to their appearance as hand samples, they contain a high abundance of hematite, resulting in NRM intensities averaging 10^{-2} A m^{-1} (10^{-5} emu/cm^3). The Zijderveld [1967] diagram in Figure 2a shows a univectorial behavior and a linear decay to the origin of a normal polarity characteristic direction. The blocking temperatures, mostly between 660° and 673°C, indicate that the hematite is relatively pure (titanium free).

Folds with a wavelength of about 1 m gave an opportunity to perform a within-site fold test. Figures 3a and 3b demonstrate that the characteristic directions improve their clustering after applying bedding corrections. The fold test is significant at the 99% confidence level [McElhinny, 1964]. The resultant mean direction from 10 samples is D/I = 316.7°/-38.4° (Table 1). Since this direction is different from directions observed in Devonian and younger rocks of the North American craton, the gray sandstones are unlikely to be remagnetized. The characteristic direction, moreover, is consistent with the bedding-corrected direction obtained from the Rose Hill dolomitic sandstones. Because of the mutual consistency combined with a positive fold test, we are confident that the characteristic direction is of Silurian age.

Red sandstones. Forty-three hematitic red sandstones from Pinto were analyzed with thermal and chemical techniques, resulting in a mean characteristic direction of D/I = 169.1°/-5.0° (Table 1). Tilt corrections for these samples are all nearly identical, so there is no possibility of a within-site fold test. However, a comparison of the directions from the dolomitic sandstones and the gray sandstones with

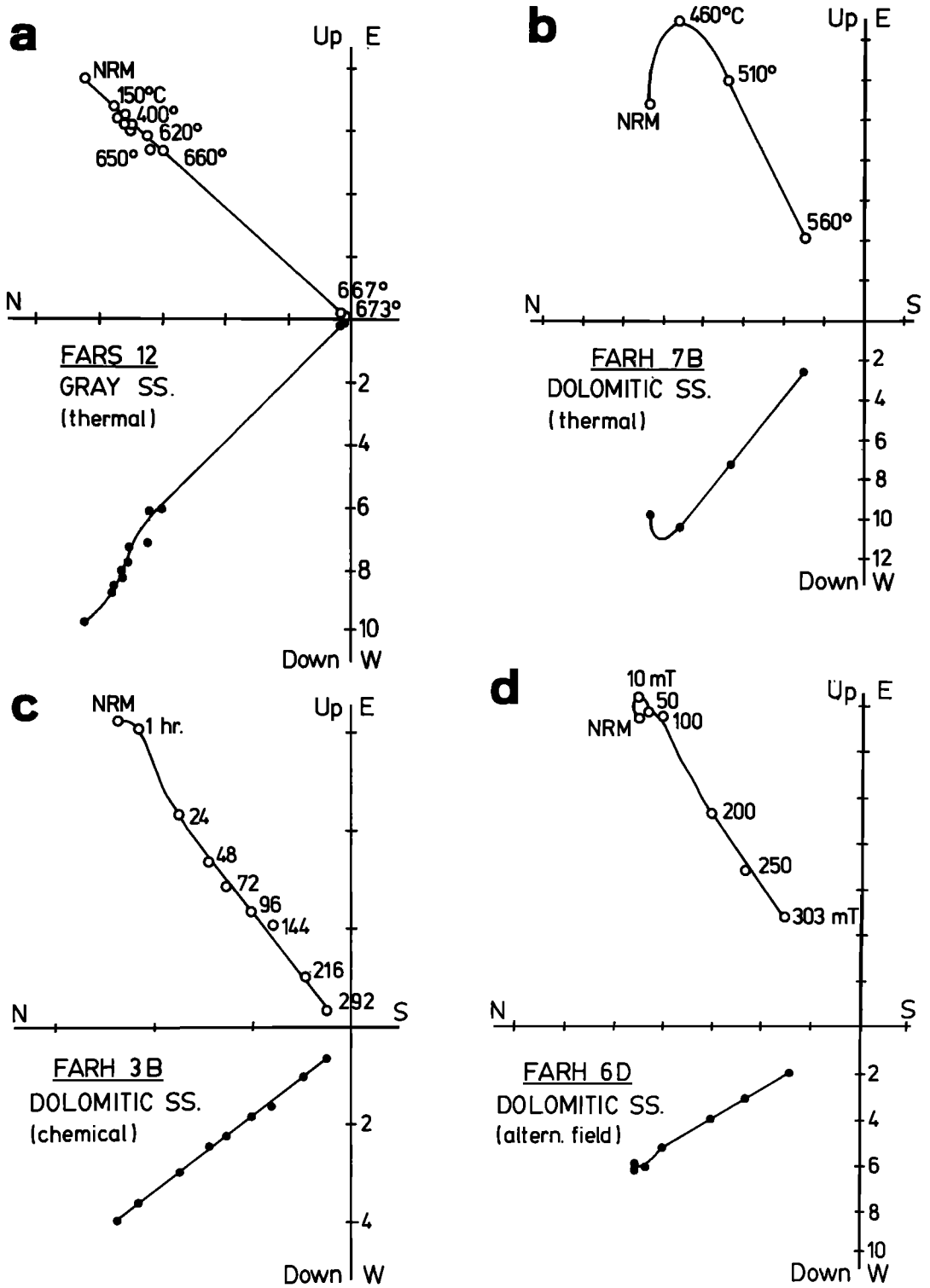


Fig. 2. Orthogonal demagnetization projections [Zijderveld, 1967] of the successive endpoints of the magnetization vector during progressive demagnetization, after correction for the tilt of the strata. Solid circles represent projections on the horizontal plane, and open symbols those on the north-south vertical plane. Units indicated for the axes are in 10^{-3} A m^{-1} (10^{-5} emu per sample of 10 cm^3). Demagnetizations of (a) a gray sandstone sample from Cacapon Mountain with thermal techniques, and of (b) dolomitic sandstone samples from Pinto with thermal, (c) chemical, and (d) alternating field techniques.

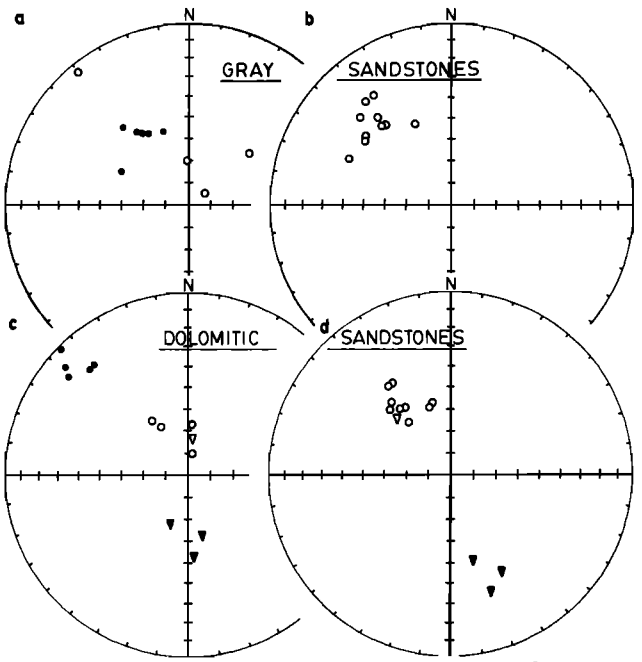


Fig. 3. Equal-area projections of the characteristic directions obtained from the gray sandstone samples (a) before and (b) after correction for the tilt of the strata and from the dolomitic sandstone samples (c) before and (d) after correction for the tilt of the strata. Open (solid) symbols represent projections on the upper (lower) hemisphere. For the dolomitic sandstones the circles denote samples from the Pinto site, and the triangles those from the Cacapon Mountain site.

those of the redbeds shows that they are distinctly different, and we conclude that the red beds acquired their magnetization in a manner unlike the adjacent sandstones. Judging from the reversed polarities and shallow inclinations, these red beds were mostly likely remagnetized by chemical processes during or after deformation in late Paleozoic times. A similar conclusion was reached by French and Van der Voo [1977]; when the studies are taken together, the reversed polarity directions of the hematitic red sandstones produce a negative fold test. On the other hand, some of the samples of the first study yielded normal polarity directions with a postive fold test (D/I = 344.1°/-42°; 12 samples from 6 sites; Table 1), after a Late Paleozoic magnetic overprinting had been removed in thermal and chemical demagnetization. In the red sandstone samples analyzed from Pinto, such normal polarity directions were only observed with any confidence in two or three samples.

Ferroan dolomitic sandstones. From the upper shale units of the Rose Hill, 13 dolomitic sandstone samples were analyzed: 9 from a small but tight fold at Pinto and 4 from a moderately dipping section at Cacapon Mountain.

The results of the thermal demagnetizations (Figure 2b) show two magnetic components: one, a small component with variable directions and low in intensity, and the other, the characteristic direction. The first component is characterized by blocking temperatures below 400°C and generally steep positive inclinations, suggesting that this is a recent magnetization. The characteristic directions have blocking temperatures generally above 450°C. Samples from Pinto are well behaved up to only 500°-560°C in thermal demagnetizations (Figure 2b). Cacapon Mountain samples show somewhat erratic thermal demagnetization patterns (Figure 4a), yet these as well as the Pinto dolomitic sandstones contain prefolding characteristic directions. The characteristic directions could

TABLE 1. Summary of the Characteristic Directions

Rock type, site, and Reference	N	Declination/ Inclination Before T.C.	k_1	α_{95}	Declination/ Inclination After T.C.	k_2	α_{95}	Pole Position
Gray sandstones, Cacapon, this study	10	333.5/+20.1	2.0	48.5	316.7/-38.4	53.5	6.7	16.6°N, 143.6°E
Dolomitic sandstones, Cacapon, this study	4	360.9/-64.4	62.2	11.7	334.9/-44.6	37.9	15.1	20.1°N, 125.6°E
Dolomitic sandstones, Pinto, this study	9	320.8/-21.3	3.0	35.6	327.6/-50.8	77.2	5.9	12.8°N, 129.1°E
Red beds, normal polarity, French and Van der Voo [1977]	12	23.2/-23.4	1.5	55.0	344.1/-42.0	10.4	14.0	25.9°N, 116.2°E
Red beds, reversed polarity this study	43	169.1/- 5.0	30.9	3.9	169.4/+ 8.4	31.1	3.9	45.1°N, 116.7°E
Mean, all samples	35	344.6/-21.1	1.9	25.8	330.6/-44.2	18.4	5.8	19.0°N, 129.1°E
Mean, all sites	9	5.2/-39.3	2.4	43.3	341.7/-44.1	26.4	10.2	22.4°N, 119.0°E
Mean, lithologies	3	350.7/-17.8	3.8	75.2	330.0/-43.7	49.7	17.7	19.1°N, 130.0°E

N is the number of samples (or other entries) given unit weight in the statistical analysis; no samples have been rejected. Declination and inclination are in degrees; T.C. denotes correction for the tilt of the strata; k and α_{95} are the precision parameter and radius of the cone of 95% confidence around the mean, respectively, with subscripts 1 and 2 denoting before and after correction for the tilt of the strata, respectively.

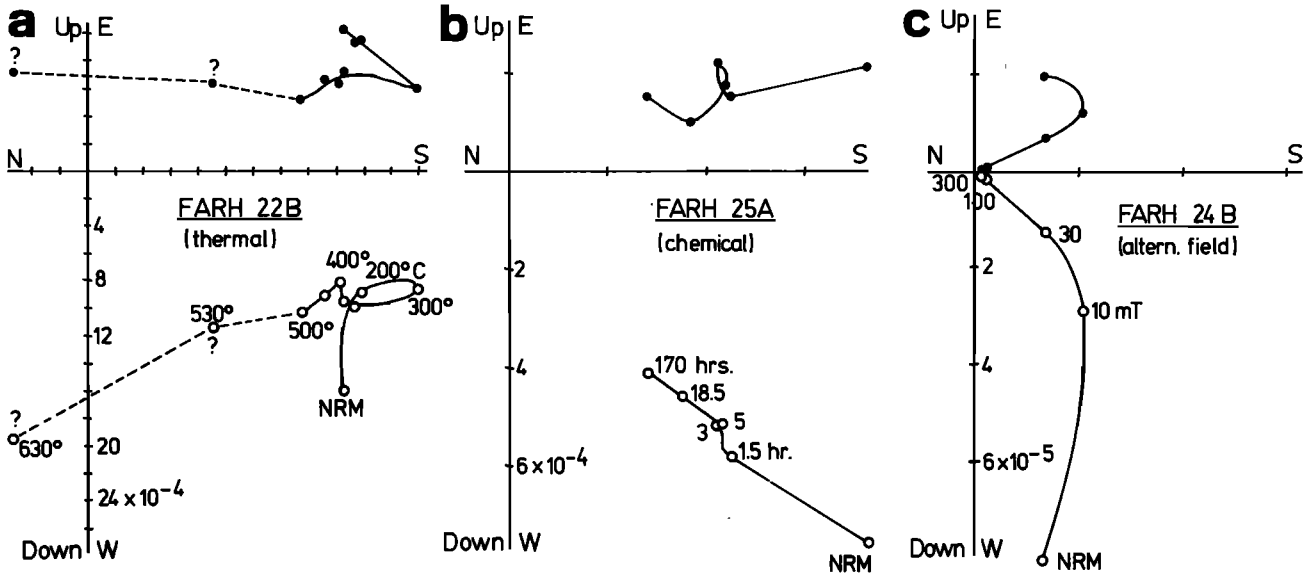


Fig. 4. Orthogonal demagnetization projections as in Figure 2 of dolomitic sandstone samples from the Cacapon Mountain site obtained with (a) thermal, (b) chemical, and (c) alternating field techniques. Note the apparent instability occurring at higher temperatures during thermal treatment. Units indicated for the axes are in amperes per meter.

be obtained from individual thermal, chemical, and alternating field demagnetizations. There was excellent directional consistency between all three methods and throughout the formation, but in thermal treatment the samples did not always reach a stable endpoint below 500°C. Above this temperature, changes in the mineralogy began to take place, most likely a breakdown and oxidation of (nonmagnetic) ferrous minerals, and often further thermal progressive demagnetization was fruitless as the samples became susceptible to spurious magnetizations (e.g., Figure 4a). This change in mineralogy will be discussed further in the next section.

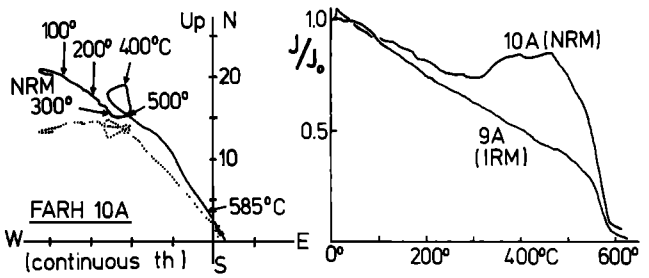


Fig. 5. Orthogonal demagnetization projection as in Figure 2 of a dolomitic sandstone sample from the Pinto site. Solid line represents projections on the horizontal plane, and dotted line those on the east-west vertical plane. Measurements were made continuously during heating. Units indicated for the axes in 10^{-3} A m^{-1}. On the right the normalized intensity is plotted versus temperature for the dolomitic sandstone sample of the left diagram (10A) as well as for a sample containing a saturation isothermal remanent magnetization (9A). Note that the loop in the diagram on the left and the corresponding increase in intensity above 320°C (on the right) are only seen in the thermal treatment of the natural remanent magnetization.

In one experiment a sample (Figure 5) was measured while being continually thermally demagnetized in the field-free space of a modified Digico-spinner configuration designed by Heiniger and Heller [1976]. The direction of the magnetization removed during this treatment is similar to the direction obtained by chemical, alternating field, and progressive thermal treatment and assures us that between 450° and 580°C we are dealing with a characteristic component of magnetization, as originally defined by Gregor and Zijderveld [1964], and not with a multicomponent remanence. Below 400°C, however, there is a loop in the demagnetization diagram which might indicate a relatively low temperature, antiparallel (about D/I = 160°/-5°) component of magnetization. We recall that this direction was also found in the red sandstones and by French and Van der Voo [1977] and has been interpreted as a synfolding to postfolding Permo-Carboniferous partial to complete remagnetization. The issue has not been completely resolved, however, since it is theoretically possible that new minerals forming in the sample may acquire a spontaneous magnetization, parallel to the magnetization present in the sample, thus causing the loop in Figure 5. Since the magnetic effects associated with the mineralogy changes (next section) were found to occur largely above 450°C, we are inclined to prefer the explanation that the loop is caused by a small remagnetization component, removed below 400°C.

Two relationships indicate the characteristic directions (Table 1) to have been stable since at least 250 m.y. ago, and probably much longer: (1) a fold test and (2) the presence of both normal and reversed polarity directions. The fold test is our strongest stability indicator. Figures 3c and 3d show the improvement in the cluster of directions after tilt corrections have been applied. This fold test is significant at the 99% confidence level

[McElhinny, 1964]. Suggestion of stability is further indicated by the recording of both normal and reversed polarities at Cacapon Mountain. This is not a stringent stability test, but we reason that if the dolomitic sandstones were susceptible to remagnetization during the Late Carboniferous/Early Permian reversed Kiaman interval, then they are unlikely to show antiparallel directions. There is room for interpretation of the Cacapon Mountain dolomitic sandstone directions owing to the difficulty in thermal demagnetizations of these weakly magnetized rocks, but alternating field demagnetization of two samples (e.g., Figure 4c) yields stable and roughly antiparallel directions at 10 mT or more of 317.6°/-54.7° and 152.0°/39.2°. This is consistent with the normal polarity directions obtained from the Pinto dolomitic sandstones, as well as with those of gray sandstones and the normal polarity red beds of French and Van der Voo [1977]. The dolomitic sandstones from both Pinto and Cacapon Mountain sites yield a mean direction of 330°/-48.9° (Table 1).

Rock Magnetic Experiments (Dolomitic Sandstones)

The dolomitic sandstones are unusual rocks; described as siderites ($FeCO_3$) in the literature, they were observed by optical microscope studies, electron microprobe and microscope studies, and powder X ray diffraction to contain ankeritic (ferroan) dolomite cement (about 45%). In addition, an iron-bearing chloritelike mineral, chamosite, was observed (about 10%). Chamosite is a diagenetic mineral, of the septechlorite or kaolinite-serpentine series, with Mg, Fe^{3+}, Fe^{2+}, and Al in the octahedral layers. Using the rocks in a paleomagnetic study created special problems because the magnetic remanence and its carriers were difficult to identify owing to severe mineralogic changes that accompanied thermal treatment. However, a series of rock magnetic experiments enabled us to identify the magnetic carriers both in the natural and in the altered (heated) states.

Only hematite and magnetite appear to contribute to the overall NRM of the Pinto and Cacapon Mountain dolomitic sandstones. The total contribution to the NRM by magnetite is

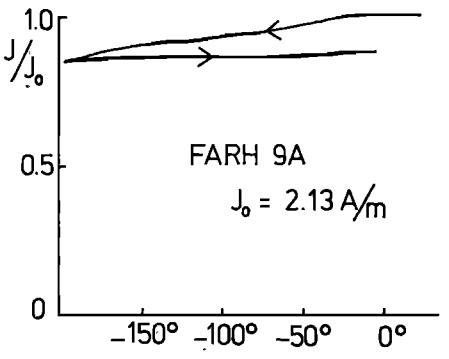


Fig. 6. Low-temperature behavior of saturation isothermal remanent magnetization acquired at room temperature. Sample from the dolomitic sandstones of the Pinto site; no clear transitions are seen, suggesting small grain sizes for the remanence-carrying minerals.

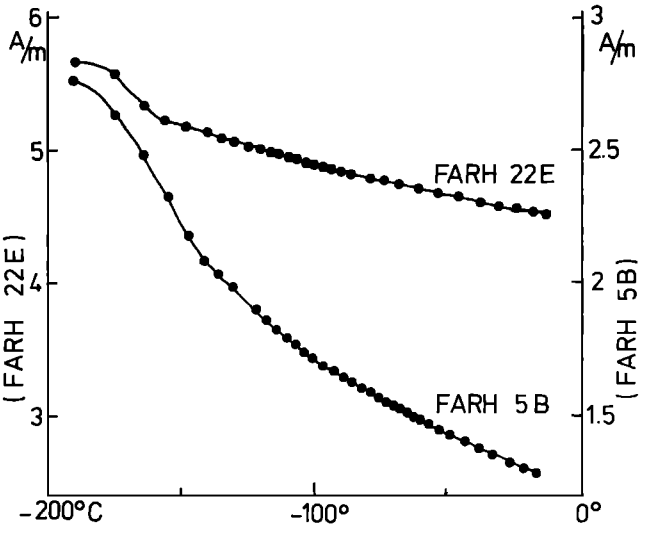


Fig. 7. Low-temperature behavior of saturation isothermal remanent magnetization acquired at -196°C. Samples from the dolomitic sandstones of Pinto (5B) and Cacapon Mountain (22E); a small but noticeable transition is seen at about -165°C, suggesting the presence of magnetite.

not known, as its properties are difficult to distinguish from low-coercivity hematite, but thin sections of both Pinto and Cacapon Mountain dolomitic sandstones show accessory amounts of detrital magnetite and only traces of hematite. Some of this magnetite in Cacapon Mountain dolomitic sandstones appears to be multidomained, because it shows large viscous remanent magnetizations (VRM's) and a removal of 80% of its initial intensity at 10 mT. The Pinto samples have greater stability and coercivities well over 100 mT. A few samples with saturation isothermal remanent magnetization (IRM) were cooled to -196°C and then reheated to room temperature (Figure 6). No Morin or magnetite transitions were observed. The latter implies dominance of shape anisotropy for magnetite, which in turn suggests that fine mineral grains are much more numerous than coarse mineral grains, insofar as they carry a remanence. When an IRM is imparted at low temperatures (Figure 7), a small but noticeable transition is seen at -160°C, indicating magnetite. A transition characteristic of hematite is not seen at -20°C, but hematite must be present, since magnetite has a theoretical maximum coercive force of 300 mT and alternating field demagnetization to that value (Figure 2d) does not remove all of the NRM. IRM imparted to these samples (Figure 8a) corroborates this, as saturation is not reached until about 1.0 tesla. Also, thermal demagnetizations did not totally erase the remanence at the magnetite Curie point of 580°C (Figure 5).

Mineralogical changes during heating of the dolomitic sandstones were also investigated. With the sharp increase in saturation IRM at about 560°C (Figure 8b), it is likely that a breakdown and oxidation of ferroan dolomite or of chamosite occurs. Comparison of thin sections taken from unheated and heated (to 600°C) dolomitic sandstones confirms that

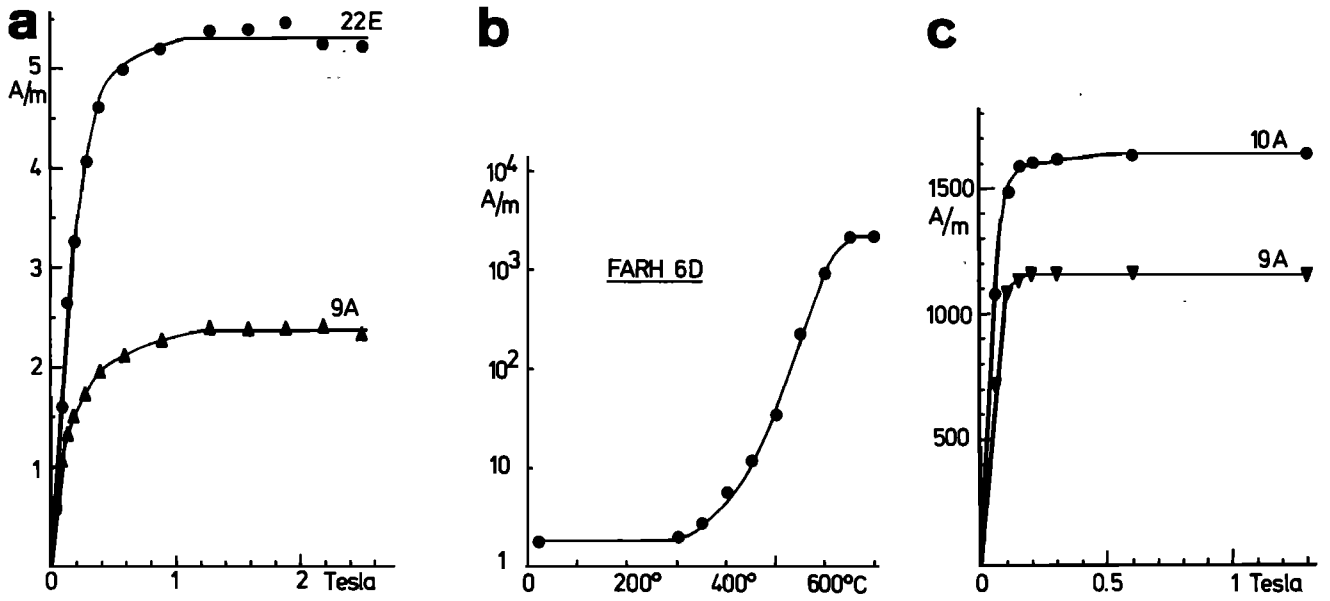


Fig. 8. Behavior of isothermal remanent magnetization (IRM), acquired at room temperature, for dolomitic sandstone samples from Pinto (9A, 6D, 10A) and Cacapon Mountain (22E). (a) IRM acquisition of unheated dolomitic sandstone, reaching saturation at about 1 tesla (10 k Oe). (b) Intensity of saturation IRM measured and induced (in 1.3 tesla) after each step in a series of heating (and cooling) steps to progressively higher temperatures. The sharp increase above 400°C in saturation IRM intensity in this semilog plot suggests that profound mineralogical changes occur during heating. (c) Acquisition of IRM by dolomitic sandstone samples which had been heated to above 600°C and then cooled. Note the increase in saturation IRM intensity of 3 orders of magnitude over the sample in Figure 8a and the fact that saturation now is rapidly reached below 0.3 tesla (3 k Oe).

opaque minerals have formed after chamosite, and these new minerals appear to make up about 5-10% of the sample. In their altered state the sandstones reached a saturation IRM of about 1.2-2.0 x 10^3 A m^{-1} (Figure 8c). Since hematite has a saturation magnetization of approximately 2.5 x 10^3 A m^{-1} (while magnetite's saturation level is at 478 x 10^3 A m^{-1}), hematite alone cannot account for all of the saturation IRM.

From this and from the fact that the altered (heated) samples rapidly reach saturation at about 0.2 tesla (Figure 8c) we conclude that fine-grained magnetite must make a substantial contribution to the isothermal remanence of the heated dolomitic sandstones. In a late stage of the investigation, newly formed magnetite was indeed observed in X ray studies after heating of a cylindrical sample.

In summary, the heating of the dolomitic sandstones appears to produce both hematite and magnetite, thereby increasing the saturation IRM by 3 orders of magnitude. The unheated samples also contain both hematite and magnetite. The larger magnetite grains were observed to be of detrital origin. The paleomagnetic direction of the magnetite is similar to that of the hematite in the dolomitic sandstones as well as in the gray sandstones.

Discussion

The preceding results lead to a reliable pole position for the middle Silurian Rose Hill Formation at 19.0°N, 129.1°E. This is based upon an average of the 35 sample directions (k = 18.4, α_{95} = 5.8°, dp = 4.6°, dm = 7.3°; Table 1). Mean directions and pole positions, giving unit weight to the nine sites or the three lithologies, are also given in Table 1 but appear to be less desirable because of the uneven distribution of normal polarity samples per site for the study of French and Van der Voo [1977]. In justification of our conclusion that the pole is reliable we further cite the following:

(1) The bedding-corrected characteristic directions from the dolomitic sandstones and gray sandstones are mutually consistent, as are those of the normal polarity red beds (Figure 9).

(2) Fold tests performed on the characteristic directions of the dolomitic sandstones, the gray sandstones, and normal polarity hematite sandstones [French and Van der Voo, 1977] are independently conclusive at the 99% confidence level. (3) The characteristic directions, from which the Rose Hill pole position is derived, are based upon detailed thermal, chemical, and alternating field demagnetizations which give consistent and repeatable results. (4) The normal-polarity directions, virtually the only ones used to calculate the pole position, are unlikely to have been the result of remagnetization during tectonism in the predominantly reversed Kiaman interval during Late Carboniferous to Early Permian times. Furthermore, they do not have the shallow inclinations typical for North America in the late Paleozoic.

We feel that any postdepositional remag-

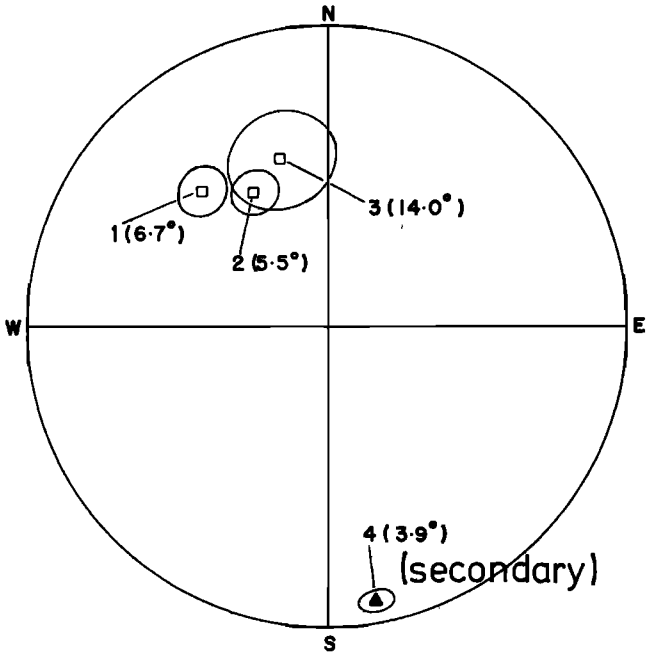


Fig. 9. Equal-area projection of the mean characteristic directions of magnetization, after correction for the tilt of the strata: 1, gray sandstones; 2, dolomitic sandstones; 3, normal polarity hematitic red sandstones [French and Van der Voo, 1977]; and 4, remagnetized hematitic red sandstones (this study). The α_{95} values are indicated in parentheses.

netizations of the Rose Hill Formation, as evident in the reversed polarity red beds (see Figure 9), must have been a chemical and not a thermal event, since any thermal effects would be likely to remagnetize all the rocks. The results show that only the red bed units have been remagnetized, while the overlying and underlying sandstones have prefolding magnetizations. Detailed optical and rock magnetization studies of the Pinto and Cacapon Mountain dolomitic sandstones indicate accessory amounts of fine-grained detrital magnetite and trace amounts of hematite. It appears that both the magnetite and the hematite grains contain the characteristic direction.

We now compare the results of this study with those obtained by Graham [1949]. The NRM's of the Pinto dolomites sampled by Graham and by us are normal polarity directions, and they are in close agreement. Reversed polarity directions from Graham's Cacapon site are not tightly clustered, and since there was a large VRM in some of our samples from Cacapon Mountain, comparison of his reversed polarity NRM values with our directions is not meaningful. After demagnetization, our mean characteristic normal polarity direction is more negatively inclined than Graham's mean NRM direction. This disagreement is a direct result of a small, soft magnetic component which is removed below 410°C. Despite Graham's unknowing inclusion of this postfolding component, his fold test passed at the 99% confidence level. The fold test is powerful, but if the direction is a multicomponent vector, it only has the potential to indicate the presence, and not the full nature, of a prefolding component. Moreover, there are instances in which a post-folding component will not even be revealed with the fold test, namely, when the fold angle is small, when the dispersion of directions from strata on each limb is large, or when the postfolding magnetization lies along the fold axis (the latter, incidentally, is not the case: the fold axes in our area strike N27°E ± 10°). To be fully sure of a fold test's significance, one must minimally consider demagnetization results as well as any other stability tests that can be applied.

Having done so, we are confident that our pole position is meaningful. Most likely, this Rose Hill pole is of Silurian age: it is significantly different from reliable Devonian and younger poles thus far obtained for North America [Irving, 1979]. The Rose Hill pole is younger than the Early Silurian Castanea pole (unpublished data quoted by McElhinny and Opdyke [1973]) and older than the Late Silurian Bloomsburg pole [Roy et al., 1967], yet it lies well to the southeast of both of them. If all three poles are valid for the North American craton and are based on primary magnetizations, a rapid looping in the apparent polar wander path of North America must have occurred. Alternatively, local relative rotations may have occurred, affecting the declinations but not the inclinations (Castanea: -44°, Rose Hill: -44°, Bloomsburg: -32°). This would simplify the North American polar wander curve and minimize elaborate wanderings and loops and would explain the declination discrepancies in the characteristic directions. Aberrant declinations could also follow from improper tilt corrections, since we did not account for rotation due to plunging folds. There does not appear to be any field evidence, however, for steeply plunging folds (>10°) or for large rotations on a scale of 1 m to 100 km. Moreover, studies of the Ordovician Juniata Formation and the Late Devonian Catskill Red beds included sites near those of our own study [Van der Voo and French, 1977; Van der Voo et al., 1979] and did not show any deviating declinations.

At this point we do not have the necessary resolution or stratigraphic coverage to decide which reconstruction is the correct one for middle Silurian times. Yet, even if one admits that the Rose Hill Formation in Maryland may have undergone a more complicated deformational history than can be recognized, one can still determine meaningful paleolatitudes. An important result from this study is that the North American continent may have been further south in Silurian times than was previously thought on the basis of the Bloomsburg result alone [Morris, 1976].

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